Find the x-intercept of the line normal to the tangent line of $y = \cos x$ at $x = \frac{\pi}{3}$

$m = -\frac{1}{-\sin\frac{\pi}{3}}$

$m = \frac{2}{\sqrt{3}}$

$\cos\frac{\pi}{3} = \frac{2}{\sqrt{3}}\frac{\pi}{3} + b$

$b = \frac{1}{2} - \frac{2\pi}{3\sqrt{3}}$

$y = \frac{2}{\sqrt{3}}x + \frac{1}{2} - \frac{2\pi}{3\sqrt{3}}$

$0 = \frac{2}{\sqrt{3}}x + \frac{1}{2} - \frac{2\pi}{3\sqrt{3}}$

$x = \frac{\pi}{3} - \frac{\sqrt{3}}{4}$